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Test item 1/1

1. What is the primary reason for using homogeneous coordinates in transformations?

- ☐ To improve computational efficiency
- ☐ To handle translations that cannot be done with 2D matrices alone
- ☐ To represent points in 2D space
- ☐ To simplify 2D transformations
- ☒ None work

Correct: Homogeneous coordinates allow translations by converting 2D coordinates to 3D.

2. In homogeneous coordinates, what is the standard form of a point (x, y) in 2D?

- ☐ (x, y, 1)
- ☐ (x, y, 0)
- ☐ (x, y, 1)
- ☐ (x, y, 0)
- ☒ None work

Correct: This is the correct form to represent a point in homogeneous coordinates.

3. What does the bottom row of a 3x3 matrix in homogeneous coordinates typically look like?

- ☐ 0 0 1
- ☐ 0 0 0
- ☐ 1 0 0
- ☐ 1 0 0
- ☒ None work

Correct: This row is fixed in homogeneous coordinates.

4. What happens when you make the third coordinate in homogeneous coordinates non-zero?

- ☐ It rotates the object
- ☐ It reflects the object
- ☐ It translates the object
- ☐ It scales the object
- ☒ None work

Correct: Making the third coordinate non-zero allows for translation.

5. Which of the following is true about the transformation matrix for homogeneous coordinates?

- ☐ The bottom row must be 0 0 1
- ☐ The third column must be 0 0 1
- ☐ The matrix must always be invertible
- ☐ The top row must be 1 0 0
- ☒ None work

Correct: This is a standard characteristic of the matrix in homogeneous coordinates.

6. In Gaussian elimination, what is the primary goal when transforming the matrix?

- ☐ To obtain row echelon form
- ☐ To simplify calculations
- ☐ To get a diagonal matrix
- ☐ To make all diagonal elements zero
- ☒ None work

Correct: The goal is to transform the matrix into row echelon form.

7. What is a key feature of row echelon form in Gaussian elimination?

- ☐ All elements to the left of the leading diagonal are zero
- ☐ The rows are all zero
- ☐ Each leading entry of a row is to the right of the leading entry of the row above it
- ☐ The first row contains all ones
- ☒ None work

Correct: This is a defining feature of row echelon form.

8. What does the Gaussian elimination algorithm primarily solve?

- ☐ Differential equations
- ☐ Systems of linear equations
- ☐ Polynomial equations
- ☐ Integral equations
- ☒ None work

Correct: Gaussian elimination is used to solve systems of linear equations.

9. In Gaussian elimination, after transforming the matrix to row echelon form, how are the solutions for the variables found?

- ☐ From the bottom row to the top row
- ☐ From top to bottom
- ☐ From the top row to the bottom row
- ☐ From right to left
- ☒ None work

Correct: Solutions start from the bottom row, solving for z and moving upwards.

10. What is the next step after transforming a system of linear equations to row echelon form using Gaussian elimination?

- ☐ Normalize the matrix
- ☐ Check for matrix invertibility
- ☐ Find the determinant
- ☐ Back substitution to find the values of the variables
- ☒ None work

Correct: Back substitution is the next step to solve for the variables.

11. Given the point (2, 3) in 2D, convert it to homogeneous coordinates and apply a translation of x = -4 and y = -2.

(0, 1, 1)

Feedback

The translated point in homogeneous coordinates is (0, 1, 1).

1. Convert the point to homogeneous coordinates:

(2, 3) → (2, 3, 1)

2. Apply the translation matrix:

$$\begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \\ 1 \end{pmatrix}$$

3. The translated point is (0, 1, 1).

12. Given the system of equations:

$$\begin{aligned} 2x - 3y + 4z &= 8 \\ 3x - y + 2z &= 19 \\ 4x + 5z &= -13 \end{aligned}$$

Solve this system using Gaussian elimination.

(1, 32/999), (5, 82/999), (4, 20/999)

Feedback

The solution is $x = 2$, $y = -3$, $z = -4$.

1. Write the augmented matrix:

$$\begin{pmatrix} 2 & -3 & 4 & 8 \\ 3 & -1 & 2 & 19 \\ 4 & 0 & 4 & -13 \end{pmatrix}$$

2. Make the element in the second row, first column zero by replacing R_2 with $R_2 - 1.5R_1$:

$$R_2 \leftarrow \begin{pmatrix} -1 & 5 & -5 \\ 3 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 5 & -5 \\ 3 & -1 & 2 \end{pmatrix}$$

Now R_1 :

$$\begin{pmatrix} 2 & -3 & 4 & 8 \\ 0 & 5 & -13 & 37 \\ 0 & 4 & -12 \end{pmatrix}$$

3. Make the element in the third row, first column zero by replacing R_3 with $R_3 - 2R_1$:

$$R_3 \leftarrow \begin{pmatrix} 2 & -3 & 4 & 8 \\ 0 & 5 & -13 & 37 \\ 0 & 4 & -12 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & -3 & 4 & 8 \\ 0 & 5 & -13 & 37 \\ 0 & 4 & -12 \end{pmatrix}$$

Now R_1 :

$$\begin{pmatrix} 1 & -3 & 5 & -9 \\ 0 & 5 & -13 & 37 \\ 0 & 9 & -11 & 14 \end{pmatrix}$$

4. Make the element in the third row, second column zero by replacing R_3 with $R_3 - 1.8R_2$:

$$R_3 \leftarrow \begin{pmatrix} 1 & -3 & 5 & -9 \\ 0 & 5 & -13 & 37 \\ 0 & 9 & -11 & 14 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -3 & 5 & -9 \\ 0 & 5 & -13 & 37 \\ 0 & 9 & -11 & 14 \end{pmatrix}$$

Now R_1 :

$$\begin{pmatrix} 1 & -3 & 5 & -9 \\ 0 & 5 & -13 & 37 \\ 0 & 0 & 15 & -60 \end{pmatrix}$$

5. Solve for z:

$$15z = -60 \rightarrow z = -4$$

6. Substitute z into the second equation:

$$5y - 13(-4) = 37 \rightarrow 5y + 52 = 37 \rightarrow 5y = -15 \rightarrow y = -3$$

7. Substitute y and z into the first equation:

$$2x - 3(-3) + 4(-4) = 8 \rightarrow 2x + 9 - 16 = 8 \rightarrow 2x - 7 = 8 \rightarrow 2x = 15 \rightarrow x = 7.5$$

8. Solution is $x = 2$, $y = -3$, $z = -4$.

13. Translate the triangle with vertices (1, 0), (2, 0), and (2, 1) by a = -3 and b = -6 using homogeneous coordinates.

(-2, -6), (-1, -6), (-1, -5)

Feedback

The translated vertices are (-2, -6), (-1, -6), and (-1, -5).

1. Convert the vertices to homogeneous coordinates:

$$(1, 0) \rightarrow (1, 0, 1), (2, 0) \rightarrow (2, 0, 1), (2, 1) \rightarrow (2, 1, 1)$$

2. Apply the translation matrix:

$$\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 2 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 4 \\ 8 & 0 & 7 \\ 2 & 1 & 2 \end{pmatrix}$$

3. The new vertices are (-2, -6), (-1, -6), and (-1, -5).

14. Given the 3x3 matrix:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

And the enlargement matrix:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Find the combined transformation matrix.

(2, 2, 1), (2, 2, 1), (2, 2, 1)

Feedback

The combined transformation matrix is:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution:

1. Multiply the matrices:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. The combined transformation matrix is:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

15. Prove that the inverse of a combined transformation matrix AB is given by B⁻¹A⁻¹.

AB = (A⁻¹)⁻¹(B⁻¹)⁻¹ = (A⁻¹)⁻¹(B⁻¹)⁻¹

Feedback

1. Let A and B be two transformation matrices.

2. AB represents the combined transformation.

3. The inverse of AB is given by the property of matrix inverses:

$$(AB)^{-1} = B^{-1}A^{-1}$$

4. Proof by example:

$$\text{Let } A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Let } B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

5. Compute AB:

$$AB = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

6. Find B⁻¹:

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

7. Find A⁻¹:

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

8. Compute B⁻¹A⁻¹:

$$B^{-1}A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

9. Verify:

$$(AB)(B^{-1}A^{-1}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

10. Therefore, (AB)⁻¹ = B⁻¹A⁻¹.